

**Problem 1.** Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$  be functions of sets.

- (a) Prove that if  $g \circ f$  is injective then  $f$  is injective.
- (b) Prove that if  $g \circ f$  is surjective then  $g$  is surjective.
- (c) Find an example of  $f : A \rightarrow B$  and  $g : B \rightarrow A$  such that  $f \circ g = \text{id}_B$  but neither  $f$  nor  $g$  is bijective.

*Proof.* **(a)** Suppose  $g \circ f$  is injective but that  $f$  is not injective. So there exists  $a, \alpha \in A$  such that  $f(a) = f(\alpha)$  but  $a \neq \alpha$ . But then  $g(f(a)) = g(f(\alpha))$  and  $a \neq \alpha$ , which contradicts that  $g \circ f$  is injective. So  $f$  must be injective.

**(b)** Suppose  $g \circ f$  is surjective but that  $g$  is not surjective. So there exists  $c \in C$  such that for all  $b \in B$ ,  $g(b) \neq c$ . But then for all  $a \in A$ ,  $g(f(a)) \neq c$ , which contradicts that  $g \circ f$  is surjective. So  $g$  must be surjective.

**(c)** Let  $A = \{a\}$ ,  $B = \{b, \beta\}$ ,  $C = \{c\}$ . Then let  $f := a \mapsto b$  and  $g := \{b, \beta\} \mapsto c$ . Then  $f$  is not surjective, and  $g$  is not injective but  $g \circ f$  is bijective via  $(g \circ f)^{-1} := c \mapsto a$ .

□

**Problem 2.** Let  $R$  be a commutative ring, and  $S \subseteq R$  be a multiplicative set.

(a) Review the definition of  $S^{-1}R$ , and prove that  $+$  and  $\cdot$  make  $S^{-1}R$  into a commutative ring with identity  $1_R/1_R$ .

(b) If  $0 \in S$ , show that  $S^{-1}R$  is the zero ring.

**Definition 1** (Localization). Let  $R$  be a commutative ring and  $S \subseteq R$  a multiplicative submonoid;  $S$  contains  $1_R$  and is closed under multiplication. The **localization**  $S^{-1}R$  of  $R$  at  $S$  is defined as follows.

$$S^{-1}R := \left\{ \frac{r}{s} \mid r \in R, s \in S \right\} / \sim \text{ where } \frac{a}{b} \sim \frac{c}{d} \iff \exists s \in S \text{ such that } s(ad - bc) = 0.$$

We define  $+$  and  $\cdot$  on  $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$  as follows:

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}.$$

Also, notice that  $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} = \frac{ca}{db} = \frac{c}{d} \cdot \frac{a}{b}$  and  $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} = \frac{cb+da}{db} = \frac{c}{d} + \frac{a}{b}$ . Commutativity is immediate.

We now briefly prove a lemma.

**Lemma 1.** If  $\circ$  is a commutative binary operation on  $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$  such that  $x \circ y \sim x \circ y'$  for any  $x$  and any  $y \sim y'$  in  $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$ , then for any  $x \sim x', y \sim y'$  in  $(x \circ y) \sim (x \circ y') \sim (x' \circ y) \sim (x' \circ y')$ . That is:  $[x]_{\sim} \circ [y]_{\sim} = [x \circ y]_{\sim}$

*Proof.*  $x \circ y \sim x \circ y'$  for any  $y \sim y' \implies y \circ x \sim y \circ x'$  if  $x \sim x'$ . Similarly,  $x' \circ y' \sim x' \circ y$  if  $y' \sim y \implies y' \circ x' \sim y' \circ x$  if  $x' \sim x$ . So then since  $x \sim x'$  and  $y \sim y'$ , we have that  $(x \circ y) \sim (x \circ y') \sim (x' \circ y) \sim (x' \circ y')$  by transitivity.  $\square$

Next prove that  $+, \cdot$  are well-defined on  $S^{-1}R$ .

**Lemma 2.**  $+, \cdot$  are well-defined on  $S^{-1}R$ .

*Proof.* (a) We prove that  $S^{-1}R$  is a commutative ring with identity 1. To begin we show that these  $+, \cdot$  are well-defined on  $S^{-1}R$ . For any  $x = \frac{a_1}{b_1} \sim \frac{a_2}{b_2} = x'$  and  $y = \frac{c_1}{d_1} \sim \frac{c_2}{d_2} = y'$  in  $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$ , there exists  $s, t \in S$  such that  $s(a_1b_2 - b_1a_2) = 0$  and  $t(c_1d_2 - d_1c_2) = 0$ . Behold.

$$\begin{aligned} x + y &= \frac{a_1}{b_1} + \frac{c_1}{d_1} = \frac{a_1d_1 + b_1c_1}{b_1d_1}, \quad x + y' = \frac{a_1}{b_1} + \frac{c_2}{d_2} = \frac{a_1d_2 + b_1c_2}{b_1d_2} \\ x \cdot y &= \frac{a_1}{b_1} \cdot \frac{c_1}{d_1} = \frac{a_1c_1}{b_1d_1}, \quad x \cdot y' = \frac{a_1}{b_1} \cdot \frac{c_2}{d_2} = \frac{a_1c_2}{b_1d_2} \\ \implies (i) \quad &t(b_1d_2(a_1d_1 + b_1c_1) - b_1d_1(a_1d_2 + b_1c_2)) = t(\cancel{a_1b_1d_1d_2} + b_1^2c_1d_2 - \cancel{a_1b_1d_1d_2} - b_1^2d_1c_2) \\ &= tb_1^2(c_1d_2 - d_1c_2) = b_1^2(t(c_1d_2 - d_1c_2)) = b_1^2 \cdot 0 \implies x + y \sim x + y' \\ \implies (ii) \quad &t(b_1d_2(a_1c_1) - b_1d_1(a_1c_2)) = t(a_1b_1c_1d_2 - a_1b_1d_1c_2) = ta_1b_1(c_1d_2 - d_1c_2) \\ &= a_1b_1(t(c_1d_2 - d_1c_2)) = a_1b_1 \cdot 0 = 0 \implies x \cdot y \sim x \cdot y' \end{aligned}$$

By Lemma 1, since  $+, \cdot$  are both commutative binary operations on  $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$  such that  $x \circ y \sim x \circ y'$  for any  $x$  and any  $y \sim y'$ , we have that  $[x]_{\sim} \circ [y]_{\sim} = [x \circ y]_{\sim}$  for  $\circ = +, \cdot$ . So  $+$  and  $\cdot$  are well-defined on  $S^{-1}R$ .  $\square$

We finally address the problem on the next page.

*Proof. (a)* We now show that  $S^{-1}R$  is a commutative ring with unity. For the sake of hygiene, we just use  $\frac{a}{b}$  to denote any  $[\frac{a}{b}]_{\sim}$  in  $S^{-1}R$ , since  $+$ ,  $\cdot$  are well-defined on it. Behold. For any  $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in S^{-1}R$ , we show each of the ring axioms:

$$(\text{Associative } +) \frac{a}{b} + \left( \frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} + \frac{cf + de}{df} = \frac{adf + bcf + deb}{bdf} = \left( \frac{a}{b} + \frac{c}{d} \right) + \frac{e}{f}$$

$$(\text{Associative } \cdot) \frac{a}{b} \cdot \left( \frac{c}{d} \cdot \frac{e}{f} \right) = \frac{a}{b} \cdot \frac{ce}{df} = \frac{ace}{bdf} = \left( \frac{a}{b} \cdot \frac{c}{d} \right) \cdot \frac{e}{f}$$

$$(\text{Commutative } +) \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} = \frac{cb + da}{db} = \frac{c}{d} + \frac{a}{b} \quad (*\text{Also shown in Definition 1*})$$

$$(\text{Commutative } \cdot) \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} = \frac{ca}{db} = \frac{c}{d} \cdot \frac{a}{b} \implies 1_{S^{-1}R} = \frac{1}{1} \quad (*\text{Also shown in Definition 1*})$$

$$\text{Also, for any } \frac{x}{s} \in S^{-1}R, s(x1 - 1x) = s(x - x) = 0, \forall s \in S, \implies \frac{x}{s} = 1_{S^{-1}R}, \forall x \in R$$

$$(1_{S^{-1}R}) \frac{1}{1} \cdot \frac{a}{b} = \frac{a}{b} \cdot \frac{1}{1} = \frac{a}{b}$$

$$(0_{S^{-1}R}) \frac{a}{b} + \frac{0}{1} = \frac{0}{1} + \frac{a}{b} = \frac{0(b) + 1a}{1b} = \frac{1a}{1b} = \frac{1}{1} \frac{a}{b} = 1_{S^{-1}R} \frac{a}{b} = \frac{a}{b}$$

$$\text{Also for any } s \in S, s^*(s(0) - (0)1) = s^*(0) = 0, \forall s^* \in S \implies \frac{0}{1} = \frac{0}{s} = 0_{S^{-1}R}, \forall s \in S$$

$$(\text{Additive Inverse}) \frac{a}{b} + \frac{-a}{b} = \frac{-a}{b} + \frac{a}{b} = \frac{-ab + ab}{bb} = \frac{0}{bb} = 0_{S^{-1}R}$$

$$(\text{Distributive}) \frac{a}{b} \cdot \left( \frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} \cdot \frac{cf + de}{df} = \frac{acf + ade}{bdf} = \frac{ac}{bd} + \frac{ae}{bf} = \left( \frac{a}{b} \cdot \frac{c}{d} \right) + \left( \frac{a}{b} \cdot \frac{e}{f} \right)$$

$$(\text{Distributive}) \left( \frac{a}{b} + \frac{c}{d} \right) \cdot \frac{e}{f} = \frac{ad + bc}{bd} \cdot \frac{e}{f} = \frac{ade + bce}{bdf} = \frac{ae}{bf} + \frac{ce}{df} = \left( \frac{a}{b} \cdot \frac{e}{f} \right) + \left( \frac{c}{d} \cdot \frac{e}{f} \right)$$

So  $S^{-1}R$  is a commutative ring with  $1_{S^{-1}R} = \frac{1}{1}$ .

**(b)** If  $0 \in S$ , then for any  $\frac{a}{b} \in S^{-1}R$ ,  $\exists 0 \in S$  such that  $0(a1 - 0b) = 0(a) = 0$  and so  $\frac{a}{b} = \frac{0}{1} = 0_{S^{-1}R}, \forall \frac{a}{b} \in S^{-1}R$ . So  $S^{-1}R = \{0_{S^{-1}R}\}$  is the zero ring.

□

**Problem 3.** Let  $R$  be a commutative ring. Recall that we say  $R$  is Noetherian if every ideal is finitely generated. Prove that  $R$  is Noetherian if and only if every chain of ideals eventually stabilizes, i.e., if

$$I_0 \subseteq I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

is a chain of ideals, then  $I_N = I_{N+1} = I_{N+k}$  for all  $k > 0$  and some  $N \in \mathbb{N}$ .

*Proof.* ( $\implies$ ) If  $R$  is Noetherian and  $I_0 \subseteq I_1 \subseteq \cdots \subseteq I_n \subseteq \cdots$  is a countable chain of ideals in  $R$ , consider  $I = \bigcup_{i \in \mathbb{N}} I_i$ . Behold.  $\forall a, b \in I$

$$(\leq_+) : a, b \in I \implies \exists i, j \in \mathbb{N} \text{ such that } a \in I_i, b \in I_j \implies \text{Without loss of generality } i \leq j$$

$$\implies a, b \in I_j \leq_+ R \implies a - b \in I_j \subseteq I \implies I \leq_+ R$$

$$(\text{Absorption}) : a \in I \implies \exists i \in \mathbb{N} \text{ such that } a \in I_i$$

$$\implies \forall r \in R, ra = ar \in I_i \subseteq I \implies \forall r \in R, \forall a \in I, ra = ar \in I.$$

So  $I$  is an ideal of  $R$ , and since  $R$  is Noetherian,  $I$  is finitely generated. So there exist  $a_1, \dots, a_n \in I$  such that  $I = \langle a_1, \dots, a_n \rangle$ . Now, for each  $i \in [n]$ ,  $a_i \in I \implies \exists k_i \in \mathbb{N}$  such that  $a_i \in I_{k_i} \subseteq I$ . Let  $N = \max\{k_1, \dots, k_n\}$ . Then  $I_{k_i} \subseteq I_N, \forall i \in [n] \implies a_1, \dots, a_n \in I_N \subseteq I = \langle a_1, \dots, a_n \rangle \implies I_N = I$ . So then

$$I_1 \subseteq \cdots \subseteq I = I_N \subseteq I_{N+1} \subseteq \cdots \subseteq I \implies I_k = I, \forall k \geq N.$$

( $\impliedby$ ) On the other hand if every countable ideal chain stabilizes, consider any ideal  $I$  of  $R$ . If  $I = \langle 0 \rangle$  or  $I = \langle 1 \rangle$  it is finitely generated. Otherwise, pick some  $a_1 \in I$ . If  $i = \langle a_i \rangle$ , it is finitely generated. Otherwise pick some  $a_2 \in I \setminus \{a_1\}$ , if  $I = \langle a_1, a_2 \rangle$  it is finitely generated. If we continue in this fashion, we build a countable chain of ideals each contained in  $I$ :

$$\langle 0 \rangle \subseteq \langle a_1 \rangle \subseteq \langle a_1, a_2 \rangle \subseteq \cdots \subseteq I$$

Since every countable ideal chain stabilizes, there exists some  $N \in \mathbb{N}$  such that  $\langle a_1, \dots, a_N \rangle = \langle a_i \mid 1 \leq i \leq k \rangle, \forall k \geq N$ . So  $I$  is finitely generated. That is, this process must terminate after picking some  $N \in \mathbb{N}$  such that  $\langle a_1, \dots, a_N \rangle = I$ .

□

**Problem 4.** Let  $R$  be a commutative ring, let  $I_j$  be ideals of  $R$ , and let  $P$  be a prime ideal of  $R$ .

- (a) Prove that if  $I_1 \cdots I_n \subseteq P$ , then  $I_j \subseteq P$  for some  $j$ . **Hint:** Use contradiction.
- (b) By (a), if  $P \supseteq \bigcap_{j=1}^r I_j$ , then  $P$  contains one of the ideals  $I_j$ . Prove or disprove: if  $P \supseteq \bigcap_{j=1}^{\infty} I_j$ , then  $P$  contains one of the  $I_j$ .

*Proof.* (a)  $I_1 \subseteq P \implies I_1 \subseteq P$ . Now assume that for some  $k \geq 1$ ,  $I_1 \cdots I_k \subseteq P \implies I_j \subseteq P$  for some  $j \in [k]$ . Next, if  $I_1 \cdots I_{k+1} \subseteq P$ , suppose that  $I_{k+1} \not\subseteq P$  and  $I_1 \cdots I_k \not\subseteq P$ . Then  $\exists (a_1 \cdots a_k) \in I_1 \cdots I_k$  and  $\exists a_{k+1} \in I_{k+1}$  such that  $(a_1 \cdots a_k), a_{k+1} \notin P$ . But then  $(a_1 \cdots a_k)a_{k+1} \in I_1 \cdots I_{k+1} \subseteq P \implies P$  is not prime, a contradiction. So either  $I_{k+1} \subseteq P$  or  $I_1 \cdots I_k \subseteq P \implies I_j \subseteq P$  for some  $j \in [k]$ . So  $I_1 \cdots I_{k+1} \subseteq P \implies I_j \subseteq P$  for some  $j \in [k+1]$ . Finally, by induction,  $I_1 \cdots I_n \subseteq P \implies I_j \subseteq P$  for some  $j \in [n]$ .

(b) We prove the finite intersection result. For any  $(a_1 \cdots a_n) \in I_1 \cdots I_n$  and then any  $k \in [n]$ , notice that  $a_1 \cdots a_n = (a_1 \cdots a_{k-1})a_k(a_{k+1} \cdots a_n) = (a_1 \cdots a_{k-1})(a_{k+1} \cdots a_n)a_k \in I_k$ . So then  $(a_1 \cdots a_n) \in I_k$ ,  $\forall k \in [n] \implies (a_1 \cdots a_n) \in \bigcap_{i=1}^n I_i \implies I_1 \cdots I_n \subseteq \bigcap_{i=1}^n I_i \subseteq P$ . Therefore, by 4(a),  $I_1 \cdots I_n \subseteq P \implies I_1 \cdots I_n \subseteq P \implies I_j \subseteq P$  for some  $j \in [n]$ .

Now consider the countably infinite collection of ideals  $\{\langle x^i \rangle \in \mathcal{P}(k[[x]]) \mid i \in \mathbb{Z}^+\}$  of  $R = k[[x]]$  for some unital commutative ring  $k$ .  $I = \bigcap_{i=1}^{\infty} \langle x^i \rangle$ . Any element of  $I$  is of the form  $rx^j$  for some  $r \in k$  and some  $j \geq 1$ . Additionally,  $rx^j \in I \subseteq \langle x^{j+1} \rangle \implies rx^j = r'x^{j+1}$  for some  $r' \in k$ . Therefore by the definition of equality in  $k[[x]]$ ,  $r = r' = 0$  and in fact  $I = \bigcap_{i=1}^{\infty} \langle x^i \rangle = \{0\}$ , a prime ideal. However,  $\langle x^i \rangle \not\subseteq \{0\}, \forall i \geq 1$ , and we have a counterexample in this intersection of ideals in  $R = k[[x]]$ .

□

**Problem 5.** Let  $R$  be a commutative ring, and let  $\text{nil}(R)$  be the set of nilpotents of  $R$ , i.e.

$$\text{nil}(R) := \{r \in R \mid r^n = 0 \text{ for some } n \in \mathbb{N}\}.$$

This is often called the nilradical of  $R$ , and you proved in the last homework that it is an ideal of  $R$ . Prove that  $\text{nil}(R)$  is contained in every prime ideal of  $R$ .

*Proof.* Let  $P$  be any prime ideal of  $R$ , and consider any  $x \in \text{nil}(R)$ . If  $x^0 = 1 = 0$ , then  $R = \{0\}$  and so  $x = 0 \in P$ . If  $x^1 = x = 0 \in P \leq_+ R$ , then  $x \in P$ . Suppose  $x^k \in P \implies x \in P$  for some  $k \geq 1$ . Then if  $x^{k+1} = x(x^k) \in P$ , since  $P$  is prime,  $x \in P$  or  $x^k \in P \implies x \in P$ . So then by induction  $x^n \in P \implies x \in P$  for all  $n \geq 1$ . Therefore,  $\forall x \in \text{nil}(R), \exists n \in \mathbb{N}$  such that  $x^n = 0 \in P \implies x \in P$ . So  $\text{nil}(R) \subseteq P$ . Thus,  $\text{nil}(R)$  is contained in any prime ideal  $P$  of  $R$ .

□

**Problem 6.** A ring is called local if it has a unique maximal ideal. Let  $R$  be a commutative local ring. Prove that if  $x \in R$ , then either  $x$  or  $1 - x$  is a unit.

*Proof.* Let  $M$  be the unique maximal ideal of  $R$ . Since  $M$  is proper, it can't contain 1 otherwise  $1r = r1 = r \in M, \forall r \in R$ . Next, if  $y \in R$  is not a unit, then  $ry = yr \neq 1, \forall r \in R$  and so  $\langle y \rangle = \{ry \in R \mid r \in R\}$  doesn't contain 1. That is,  $\langle y \rangle \subset R = \langle 1 \rangle$ . So then  $\langle y \rangle = M$  or  $\langle y \rangle \subset M$  since  $M$  is unique and maximal. In either case  $y \in M$ . So  $M$  contains all non-units of  $R$ .

Now, suppose  $x$  and  $1 - x$  are both non-units in  $R$ . Then  $x, 1 - x \in M$ . But then  $x + 1 - x = 1 \in M$ , a contradiction. So either  $x$  or  $1 - x$  is a unit in  $R$ .  $\square$

**Problem 7.** Let  $R$  be a commutative ring, and let  $I \subseteq R$  be a proper ideal. Prove that the set of prime ideals containing  $I$  has minimal elements. These are the minimal primes of  $I$ . **Hint:** Use Zorn's Lemma.

*Proof.* Let  $S$  be the set of prime ideals of  $R$  that contain  $I$ . In **Homework 2** we proved that the set of all proper ideals containing  $I$  is a poset with set inclusion. So then  $(S \setminus \{R\}, \subseteq)$  is an induced subposet of that poset. Obviously,  $P \subseteq R, \forall P \in S \setminus \{R\}$  and so  $(S, \subseteq)$  is also a poset. Now consider any chain  $C$  in  $S$ .

Let  $L_C = \bigcap_{P \in C} P$ . Obviously,  $L \subseteq C, \forall P \in C \implies I \subseteq L$ . Next,  $\forall a, b \in L, a, b \in P, \forall P \in C \implies a - b \in P, \forall P \in C$  since they are all additive subgroups of  $R$ . So  $L \leq_+ R$ . Also,  $\forall r \in R, \forall a \in L, ra \in P, \forall P \in C \implies ra = ar \in L$ . So  $L$  is an ideal of  $R$  containing  $I$ . Finally, if  $ab \in L$ , then  $ab \in P, \forall P \in C \implies a$  or  $b \in P, \forall P \in C \implies a$  or  $b \in L$ . So  $L$  is a prime ideal of  $R$  containing  $I$ , and a lower bound for  $C$  in  $S$ . So every chain in  $S$  has a lowerbound.

So then in the dual poset  $S^*$  every chain has an upperbound in  $S$ , and by Zorn's Lemma, there exists a maximal element in  $S^*$ . That is, there exists a minimal element in  $S$ .  $\square$

**Problem 8.** Let  $\mathbb{K}$  be a field. Prove that  $\mathbb{K}$  contains a unique smallest subfield  $\mathbb{k}$  and that  $\mathbb{k}$  is isomorphic to either  $\mathbb{Q}$  or  $\mathbb{Z}/p\mathbb{Z}$  for some prime  $p$ .  $\mathbb{k}$  is called the prime subfield of  $F$ . **Hint:** Intersect all subfields of  $\mathbb{K}$ .

*Proof.* Let  $\mathcal{S}(K)$  be the set of all subfields of  $\mathbb{K}$  and  $\mathbb{k} = \bigcap_{\mathbb{F} \in \mathcal{S}(K)} \mathbb{F}$ . Obviously,  $\mathbb{F} \subseteq \mathbb{K}$ ,  $\forall \mathbb{F} \in \mathcal{S}(K) \implies \mathbb{k} \subseteq \mathbb{K}$ . We show  $\mathbb{k}$  is a subfield of  $\mathbb{K}$ .  $\forall x, y \in \mathbb{k}$ ,  $x, y \in \mathbb{F}$ ,  $\forall \mathbb{F} \in \mathcal{S}(K)$  and

$$x + y \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies x + y \in \mathbb{k}$$

$$xy \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies xy \in \mathbb{k}$$

$$-x \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies -x \in \mathbb{k}$$

$$x^{-1} \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies x^{-1} \in \mathbb{k}$$

Then,  $\mathbb{k} = \bigcap_{\mathbb{F} \in \mathcal{S}(K)} \mathbb{F} \subseteq \mathbb{F}$ ,  $\forall \mathbb{F} \in \mathcal{S}(K) \implies \mathbb{k}$  is the smallest subfield of  $\mathbb{K}$ . Well, for any other minimal element  $\mathbb{k}' \in \mathcal{S}(\mathbb{K})$ ,  $\mathbb{k}' \subseteq \mathbb{k}$  and  $\mathbb{k} \subseteq \mathbb{k}' \implies \mathbb{k} = \mathbb{k}'$ . So  $\mathbb{k}$  is the unique smallest subfield of  $\mathbb{K}$ . Next, let

$$n_+ = \sum_{i=1}^n 1_{\mathbb{k}}, \forall n \in \mathbb{Z}^+ \text{ and } 0_+ = 0.$$

If  $\text{Char } \mathbb{k} = 0$ , then let  $\phi : \mathbb{Q} \rightarrow \mathbb{k}$  be defined via  $\phi(\frac{a}{b}) = \frac{a_+}{b_+} = a_+ b_+^{-1}$ . This is well-defined since (i)  $b \neq 0 \implies b_+ \neq 0_{\mathbb{k}} \implies b_+^{-1} \in \mathbb{k}$ . Also, (ii)  $\frac{a}{b} = \frac{c}{d} \implies ad = bc \implies a_+ d_+ = b_+ c_+ \implies \frac{a_+}{b_+} = \frac{c_+}{d_+}$ . We show this is a ring homomorphism. For any  $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ ,

$$\phi(1) = 1_+$$

$$\phi\left(\frac{a}{b} + \frac{c}{d}\right) = \phi\left(\frac{ad + bc}{bd}\right) = \frac{(ad + bc)_+}{(bd)_+} = \frac{a_+ d_+ + b_+ c_+}{b_+ d_+} = \frac{a_+ d_+}{b_+ d_+} + \frac{b_+ c_+}{b_+ d_+}$$

$$= \frac{a_+}{b_+} \cdot \frac{d_+}{d_+} + \frac{c_+}{d_+} \cdot \frac{b_+}{b_+} = \phi\left(\frac{a}{b}\right) + \phi\left(\frac{c}{d}\right)$$

$$\phi\left(\frac{a}{b} \cdot \frac{c}{d}\right) = \phi\left(\frac{ac}{bd}\right) = \frac{(ac)_+}{(bd)_+} = \frac{a_+ c_+}{b_+ d_+} = \left(\frac{a_+}{b_+}\right) \left(\frac{c_+}{d_+}\right) = \phi\left(\frac{a}{b}\right) \cdot \phi\left(\frac{c}{d}\right)$$

Well,  $\frac{a_+}{b_+} = \frac{c_+}{d_+} \implies a_+ d_+ = b_+ c_+ \implies ad = bc \implies \frac{a}{b} = \frac{c}{d}$ , so  $\phi$  is injective and  $\ker \phi = \{0\}$ . So then by the **First Isomorphism Theorem**,  $\mathbb{Q} \cong \mathbb{Q}/\{0\} \cong \phi(\mathbb{k}) \subseteq \mathbb{k}$ . Well, since  $\phi(\mathbb{k}) \subseteq \mathbb{k} \subseteq \mathbb{K}$  must then be a field,  $\phi(\mathbb{k}) = \mathbb{k}$  since otherwise  $\mathbb{k}$  isn't minimal. So  $\mathbb{Q} \cong \mathbb{k}$ .

If  $\text{Char } \mathbb{k} = p > 0$ , then consider  $\mathbb{k}_p = \langle 1_+ \rangle_+ \leq_+ \mathbb{k}$ . Obviously, since  $\text{Char } \mathbb{k} = p$ ,  $\mathbb{k}_p = \{0, 1_+, 2_+, \dots, (p-1)_+\}$ . All non-zero elements of  $\mathbb{k}_p$  are units, and so  $\mathbb{k}_p \subseteq \mathbb{k} \subseteq \mathbb{K}$  is a field  $\implies \mathbb{k}_p = \mathbb{k}$  since  $\mathbb{k}$  is minimal. Well, then obviously  $\phi := n_+ \mapsto n$  is a bijection from  $\mathbb{k}$  to  $\mathbb{Z}_p$ . We show this is a ring homomorphism. For any  $a_+, b_+ \in \mathbb{k}$ ,

$$\phi(1_+) = 1$$

$$\phi(a_+ + b_+) = a + b = \phi(a_+) + \phi(b_+)$$

$$\phi(a_+ b_+) = ab = \phi(a_+) \phi(b_+)$$

So then  $\phi$  is a field isomorphism and  $\mathbb{k} \cong \mathbb{Z}_p$ .

□

**Problem 9.** Let  $P$  be a prime ideal of a commutative ring  $R$ , and let  $R_P$  be the localization of  $R$  at  $P$ .

- (a) Prove that if  $f \notin P$ , then  $f/1$  is a unit in  $R_P$ .  
 (b) Let  $I \subseteq P$  be an ideal of  $R$ . Prove that

$$IR_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \right\}$$

is a proper ideal of  $R_P$ .

- (c) Prove that if  $I$  is an ideal of  $R$  and  $I \not\subseteq P$ , then  $IR_P = R_P$ .  
 (d) Prove that  $R_P$  is a local ring, and describe its maximal ideal.

*Proof.* **(a)** For a prime ideal  $P$  of  $R$ , the localization of  $R$  at  $P$  is  $R_P = (R \setminus P)^{-1}R$  and just note that these fractions are equivalence classes. Now, if  $f \notin P$  then  $\frac{f}{f} \in R_P$  and so  $\frac{f}{f} \cdot \frac{1}{f} = \frac{f}{1} \cdot \frac{1}{f} = \frac{1}{1} = 1_{R_P}$ . So  $\frac{f}{1}$  is a unit in  $R_P$ .

**(b)** If  $I$  is an ideal of  $R$ , Consider any  $\frac{a}{b}, \frac{c}{d} \in IR_P$ .  $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$ . Suppose  $bd \in P$ . But then since  $P$  is prime,  $b \in P$  or  $d \in P$ , a contradiction. So  $bd \notin P$ . Also,  $ad-bc \in I$  since  $I$  is an ideal of  $R$ . So then  $\frac{ad-bc}{bd} \in IR_P \leq_+ R_P$ . Now, for any  $\frac{r}{q} \in R_P$  and any  $\frac{i}{b} \in IR_P$ ,  $\frac{i}{b} \cdot \frac{r}{q} = \frac{r}{q} \cdot \frac{i}{b} = \frac{ri}{qb}$ . Once more,  $qb \in P \implies q \in P$  or  $b \in P$ , a contradiction. So  $qb \notin P$  and since  $ri \in I$ ,  $\frac{ri}{qb} \in IR_P$  which must then be an ideal of  $R_P$ .

If  $I \subseteq P$ ,  $IR_P$  is not proper. Then  $1_{R_P} = \frac{1}{1} \in IR_P \implies 1 \in I \subseteq P$ . But then  $1 \notin (R \setminus P)$ , and so  $\frac{1}{1} \notin R_P$  in the first place, a contradiction. So  $I \not\subseteq P \implies IR_P$  is a proper ideal of  $R_P$ .

**(c)** We simply prove the other direction ( $\supseteq$ ) continuing after proving  $IR_P \subseteq R_P$  is an ideal in the first part of **(b)**, with the added assumption that  $I \not\subseteq P$ . Since any  $i \in I$  does not belong to  $P$ , for any  $\frac{i}{q} \in IR_P$ ,  $\frac{q}{i} \in R_P \implies \frac{q}{i} \cdot \frac{i}{q} = \frac{q}{q} \cdot \frac{i}{i} = \frac{1}{1} = 1_{R_P} \in IR_P$  since it is an ideal of  $R_P$  and so  $IR_P = R_P$ .

**(d)** Consider any proper ideal  $I_P \subset R_P$ . Since it can't contain  $1_{R_P}$ , none of its elements can be units. That is, they can't be of the form  $\frac{f}{1}$  where  $f \notin P$  by **(a)**. So then

$$I_P \subseteq \left\{ \frac{a}{b} \in R_P \mid a \in P \right\} \implies I_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \subseteq P \right\}.$$

Well, since  $I_P$  is an ideal, for any such  $\frac{a}{b} \in I_P$ , and any  $\frac{r}{q} \in R_P$ , we must have that  $\frac{a}{b} \cdot \frac{r}{q} = \frac{r}{q} \cdot \frac{a}{b} = \frac{ra}{qb} \in I_P$  and that  $\frac{a}{b} - \frac{r}{q} = \frac{aq-br}{bq} \in I_P$ . Which means that  $ra = ar \in I, \forall r \in R, \forall a \in I$  and that  $\frac{a}{b} - \frac{r}{q} \in I_P, \forall r \in R, \forall q \in R \setminus P, \forall r \in R, \forall b, q \in R \setminus P, \forall a \in I \implies \frac{a}{1} - \frac{b}{1} = \frac{a-b}{1} \in I_P, \forall a, b \in I \implies a - b \in I, \forall a, b \in I \implies I \leq_+ R$ . That is,  $I \subseteq P$  is an ideal of  $P$ . So any proper ideal  $I_P \subset R_P$  is of the form  $I_P = IP_P$  for some ideal  $I \subseteq P$ .

Well,

$$IR_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \subseteq P \right\} \subseteq \left\{ \frac{a}{b} \in R_P \mid a \in \bigcup_{I \subseteq P} I \right\} = PR_P \text{ for any ideal } I \subseteq P \text{ of } R.$$

So  $PR_P$  contains all proper ideals of  $R_P$ . Thus,  $R_P$  is a local ring with the unique maximal ideal  $PR_P$ . □



**Problem 10.** If  $R$  is a Noetherian UFD, prove that any countable descending chain of prime ideals stabilizes.

*Proof.* Consider any chain

$$P_0 \supseteq P_1 \supseteq \cdots \supseteq P_n \supseteq \cdots$$

of prime ideals in  $R$ . Since  $R$  is a UFD, each prime ideal is not principal and its generators have a prime decomposition. Next, we have to prove any descending chain of prime ideals stabilizes this is not given and does not follow from the fucking ascending chain theorem.

Now,  $P_0 = \langle x_0, \dots, x_n \rangle$  for some  $x_i \in R$

□